

THE FEDERAL POLYTECHNIC, ADO EKITI, EKITI STATE.

PROJECT REPORT

ON

**ARTIFICIAL INTELLIGENCE POWERD EDUCATIONAL SERVICE
ASSISTANT FOR NIGERIA TERTIARY INSTITUTIONS**

(THE FEDERAL POLYTECHNIC ADO EKITI AS CASE STUDY)

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CERTIFICATION

I hereby certify that this project titled “ARTIFICIAL INTELLIGENCE POWERD EDUCATIONAL SERVICE ASSISTANT FOR NIGERIA TERTIARY INSTITUTIONS (THE FEDERAL POLYTECHNIC ADO EKITI AS CASE STUDY) “was carried out by AJALA JOSHUA OLUWAFERANMI matric no FPA/CS/24/3-0089 in the department of SOFTWARE AND WEB DEVELOPMENT FEDERAL POLYTECHNIC ADO EKITI in partial fulfillment of the requirements for the award HIGHER NATIONAL DIPLOMA.

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DEDICATION

I dedicate this project to God Almighty for His guidance and strength throughout this research. I also dedicate it to my parents, family, lecturers, and friends for their continuous support and encouragement.

ACKNOWLEDGEMENT

I express my sincere gratitude to God Almighty for granting me the wisdom, knowledge, and good health to complete this research. My appreciation goes to my supervisor for invaluable guidance, constructive criticism, and encouragement throughout this study. I also thank the lecturers in my department, my family, friends, and everyone who contributed directly or indirectly to the successful completion of this project.

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List of Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

NLP – Natural Language Processing

IoT – Internet of Things

ICT – Information and Communication Technology

GDP – Gross Domestic Product.

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Education is universally recognized as one of the most important instruments for national development, economic growth, technological advancement, human capital development, and social transformation. Through education, individuals acquire knowledge, skills, competencies, values, and attitudes required to contribute meaningfully to society. In the twenty-first century, however, the education sector has undergone significant transformation as a result of rapid developments in Information and Communication Technology (ICT). Digital technologies have changed the way educational institutions conduct teaching and learning, manage administrative activities, communicate with stakeholders, store information, conduct research, and provide academic and support services. The increasing digitalization of education has therefore created opportunities for institutions to improve efficiency, accessibility, quality, and responsiveness in the delivery of educational services (UNESCO, 2023).

One of the most significant technological developments influencing contemporary education is Artificial Intelligence (AI). Artificial Intelligence refers broadly to the capability of computer-based systems to perform tasks associated with human intelligence, including learning, reasoning, problem-solving, language processing, perception, and decision-making (Russell & Norvig, 2021). AI comprises several technologies and techniques, including Machine Learning (ML), Natural Language Processing (NLP), Computer Vision, Robotics, Expert Systems, Speech Recognition, and generative AI or Large Language Models (LLMs). The rapid development of these technologies has resulted in their application across numerous sectors, including healthcare, banking, manufacturing, agriculture, transportation, government, business, and education.

The emergence of generative AI has further increased the relevance of artificial intelligence in

education. Generative AI systems are capable of producing text, images, computer code, summaries, explanations, and other forms of content based on user instructions. UNESCO (2023) observes that generative AI is developing rapidly and has the potential to support curriculum development, teaching, learning, research, and other educational activities. At the same time, its adoption raises important concerns relating to privacy, academic integrity, bias, accuracy, transparency, equity, and the responsible use of technology. Consequently, educational institutions need to approach AI adoption from both technological and human-centred perspectives.

The educational sector has become one of the major areas in which Artificial Intelligence is being explored and implemented. Universities and other tertiary institutions increasingly employ AI-powered systems to support teaching, learning, student assessment, academic advising, information retrieval, student records management, admissions, library services, and institutional decision-making. Zawacki-Richter et al. (2019), in a systematic review of AI applications in higher education, identified several areas in which AI has been applied, including assessment and evaluation, adaptive systems and personalization, profiling and prediction, intelligent tutoring systems, and administrative support. This indicates that the application of AI in higher education extends beyond classroom instruction to include institutional and administrative functions.

One particularly important application of AI in tertiary institutions is the development of **AI-powered chatbots and virtual assistants**. These systems use Natural Language Processing and related AI techniques to interact with users in conversational language. They can answer frequently asked questions, provide institutional information, direct users to appropriate departments, assist with admission enquiries, provide information about registration and fees, and support students in accessing academic and administrative services. Recent research has demonstrated that conversational AI can be useful in university environments because it

provides rapid responses and can potentially reduce the workload associated with repetitive enquiries (Dinh & Tran, 2023). Similarly, a systematic review of administrative chatbot applications in higher education found that chatbots have been applied to areas such as admissions, enrolment, student services, and financial aid, while also identifying privacy, multilingual capability, and evaluation as continuing challenges (Poriye et al., 2025).

The relevance of AI to higher education is particularly important because students interact with tertiary institutions through numerous administrative processes throughout their academic journey. These processes may include admission enquiries, course registration, payment of school fees, examination information, hostel applications, library services, transcript requests, SIWES-related information, academic calendars, departmental enquiries, and graduation procedures. When information about these services is not readily available, students may experience delays, confusion, unnecessary movement between offices, and difficulty obtaining accurate information. An AI-powered educational support system can potentially provide students with a centralized channel through which they can obtain relevant information quickly and conveniently.

Furthermore, AI can support educational institutions by automating repetitive and routine enquiries. Administrative personnel in tertiary institutions frequently respond to questions that are similar or identical, particularly during admission, registration, examination, and graduation periods. An intelligent chatbot can handle a significant proportion of such routine enquiries, thereby allowing administrative personnel to devote more time to complex cases requiring human judgement. The use of AI in this manner does not necessarily imply the complete replacement of human administrators; rather, it can complement existing administrative structures and improve the efficiency of service delivery. The broader literature on AI in higher education suggests that AI can contribute to institutional efficiency while human oversight remains important for sensitive and complex decisions (Crompton & Burke,

2023; Zawacki-Richter et al., 2019).

In the Nigerian context, the adoption of emerging technologies in higher education is becoming increasingly important because of the need to strengthen the quality, relevance, innovation capacity, and global competitiveness of tertiary institutions. The National Universities Commission (NUC) has continued to emphasize technological innovation, research, entrepreneurship, and the transformation of the Nigerian university system. In 2026, the Commission's proposed STEM Transformation Strategy specifically identified emerging technologies such as Artificial Intelligence, Robotics, Biotechnology, Advanced Manufacturing, and Renewable Energy as important to Nigeria's future economic development and competitiveness (National Universities Commission [NUC], 2026).

Despite these developments, the effective use of AI in education requires appropriate institutional policies, infrastructure, technical expertise, data protection mechanisms, and human capacity. UNESCO (2023) emphasizes that educational institutions must consider privacy, equity, ethical use, transparency, and human agency when adopting generative AI technologies. Similarly, UNESCO's AI competency framework emphasizes that users of AI should develop the ability to understand, apply, and critically evaluate AI systems rather than simply depend on them (Miao et al., 2024).

Therefore, the development of an AI-powered educational support system for tertiary institutions can be viewed as an important technological intervention aimed at improving access to institutional information and enhancing administrative support services. Such a system can provide students and other users with timely responses to frequently asked questions, reduce dependence on physical visits to administrative offices, improve communication between institutions and students, and contribute to the digital transformation of tertiary education.

The proposed system is consequently designed around the idea that Artificial Intelligence can

serve as an intelligent interface between students and educational institutions. Rather than requiring students to search through multiple sources or physically visit different offices for routine information, the system can provide a centralized conversational platform through which users can submit enquiries and receive relevant responses. Such a system, when appropriately designed with reliable institutional information and human oversight, has the potential to improve the accessibility, efficiency, and responsiveness of educational support services.

1.2 Statement of the Problem

The rapid advancement of Information and Communication Technology (ICT) has significantly transformed the operations of educational institutions worldwide. Universities, polytechnics, colleges of education, and other tertiary institutions increasingly rely on digital technologies to support teaching, learning, research, student records management, communication, and administrative activities. However, technological adoption has not completely eliminated the challenges associated with the provision of timely, accurate, and accessible administrative information to students and other stakeholders.

One of the major challenges experienced by many tertiary institutions is the continued reliance on manual or semi-automated methods of providing administrative information. Students may be required to visit different offices or contact individual members of staff to obtain information relating to admission procedures, course registration, school fees, examination schedules, hostel accommodation, SIWES placement, transcript processing, library services, academic calendars, and graduation requirements. When such information is distributed across different offices, notice boards, websites, social media platforms, or individual staff members, students may find it difficult to identify the correct source of information.

The situation can become more problematic during periods of high administrative activity. Admission periods, course registration, fee payment deadlines, examinations, result

processing, and graduation exercises often generate large numbers of enquiries. Administrative officers may therefore receive repeated questions from students concerning deadlines, procedures, requirements, and institutional policies. Responding manually to every enquiry can consume considerable time and human resources. In addition, students who are unable to reach the appropriate officer may experience delays in obtaining information or may depend on information obtained from unofficial sources.

Another major problem is the limited availability of immediate support outside normal administrative working hours. Traditional administrative support generally depends on the availability of staff during designated working periods. Consequently, a student who requires information at night, during weekends, or outside office hours may have to wait until the appropriate office becomes available. This limitation is inconsistent with the expectations of modern digital users, who increasingly expect information and services to be accessible whenever they are needed.

The problem is further compounded by the possibility of inconsistent or outdated information. Where students obtain information from multiple individuals or unofficial communication channels, differences in responses may occur. Students may consequently receive conflicting information concerning registration requirements, deadlines, procedures, or institutional services. Such inconsistencies can create confusion and may result in students missing important deadlines or following inappropriate procedures. AI-powered conversational systems can potentially address some of these challenges by providing users with information from a centralized and institutionally approved knowledge base.

1.3 Aim and Objectives of the Study

1.3.1 Aim of the Study

The main aim of this study is to develop an AI-Powered Educational Service Assistant that enhances educational support services in Nigerian tertiary institutions using The Federal Polytechnic Ado-Ekiti as a case study.

1.3.2 Objectives of the Study

The specific objectives of this study are to:

1. examine the current methods used in providing educational support services in Nigerian tertiary institutions.
2. identify the challenges associated with existing student support and information service systems.
3. examine the concepts and technologies underlying Artificial Intelligence and intelligent educational assistants.
4. design a framework for an AI-powered educational service assistant suitable for The Federal Polytechnic Ado-Ekiti.
5. identify the major AI technologies required for implementing the proposed system, including Natural Language Processing, Machine Learning, Large Language Models, and Retrieval-Augmented Generation.

1.4 Significance of the Study

This study is significant because it contributes to the growing body of knowledge on the application of Artificial Intelligence in higher education and demonstrates how intelligent technologies can improve administrative efficiency and student support services in Nigerian tertiary institutions.

The findings of this study will benefit students by providing insight into how AI-powered educational assistants can offer instant, accurate, and personalized responses to academic and administrative enquiries. Students will be able to obtain information regarding admissions,

registration, examinations, school fees, library services, hostel allocation, SIWES, and academic regulations without unnecessary delays.

1.5 Scope of the Study

This study focuses on the design and application of an AI-Powered Educational Service Assistant for Nigerian tertiary institutions, using The Federal Polytechnic Ado-Ekiti as the case study. The research examines how Artificial Intelligence technologies can be utilized to improve the delivery of educational support services, enhance communication between the institution and its stakeholders, and increase the efficiency of administrative operations.

The study is limited to the use of AI in providing non-academic administrative support rather than replacing classroom teaching or academic assessment. The proposed educational service assistant is designed to provide intelligent responses to frequently asked questions and institutional enquiries relating to admissions, school fees, course registration, examination schedules, academic calendars, hostel allocation, library services, SIWES, student records, ICT support, graduation requirements, and other general institutional services.

1.6 Limitations of the Study

Like many research studies, this project encountered certain limitations that may influence the extent of its findings.

One major limitation is the limited availability of institution-specific data concerning student enquiries, administrative workflows, and AI implementation within The Federal Polytechnic Ado-Ekiti. Consequently, some discussions are based on existing literature and generally accepted administrative practices in Nigerian tertiary institutions.

Financial and time constraints also limited the scope of the research. The study was conducted within the academic period allocated for the National Diploma project, making it impossible to undertake large-scale field implementation or extensive user testing of the proposed system.

1.7 Methodology of the Study

This study adopted a **descriptive research methodology**, which relies primarily on qualitative analysis and an extensive review of existing literature related to Artificial Intelligence and educational technology.

Relevant information was collected from textbooks, peer-reviewed journals, conference proceedings, government publications, institutional reports, reputable websites, and publications from international organizations concerned with Artificial Intelligence and education.

The methodology involved the following stages:

i. Problem Identification:

The existing challenges associated with educational support services in Nigerian tertiary institutions were identified through literature review and analysis of institutional administrative processes.

ii. Literature Review:

Relevant academic publications, books, journals, and conference papers on Artificial Intelligence, educational chatbots, Natural Language Processing, Machine Learning, and educational information systems were reviewed.

iii. System Analysis:

Existing methods of providing educational support services were examined to determine their strengths, weaknesses, opportunities, and limitations.

1.8 Definition of Terms

For the purpose of this study, the following terms are defined:

Artificial Intelligence (AI):

A branch of computer science concerned with developing computer systems capable of

performing tasks that normally require human intelligence, such as reasoning, learning, problem-solving, and language understanding.

Educational Service Assistant:

An intelligent software application designed to provide information, guidance, and administrative support to students, staff, and other stakeholders within an educational institution.

Chatbot:

A computer program that interacts with users through text or voice conversations to answer questions, provide assistance, and perform specific tasks automatically.

Natural Language Processing (NLP):

A field of Artificial Intelligence that enables computers to understand, interpret, process, and generate human language.

Machine Learning (ML):

A branch of Artificial Intelligence that enables computer systems to learn from data and improve their performance without being explicitly programmed.

Large Language Model (LLM):

An advanced Artificial Intelligence model trained on large amounts of textual data to understand and generate human-like language.

Retrieval-Augmented Generation (RAG):

An AI technique that combines information retrieval from trusted knowledge sources with language generation to produce accurate and context-specific responses.

Knowledge Base:

A structured collection of verified institutional information used by an AI assistant to provide accurate responses to users.

Cloud Computing:

The delivery of computing resources such as storage, processing power, databases, and software over the internet rather than through local computers.

Educational Technology (EdTech):

The application of technological tools and digital innovations to improve teaching, learning, research, and educational administration.

Student Information System (SIS):

A computerized platform used by educational institutions to manage student records, registration, academic performance, and administrative information.

Case Study:

A research approach that involves an in-depth investigation of a specific institution, organization, or phenomenon to obtain detailed understanding and practical insights

1.9 Organization of the Report

This project is organized into four chapters to ensure a logical and systematic presentation of the research.

Chapter One introduces the study by presenting the background, statement of the problem, aim and objectives, significance, scope, limitations, methodology, definition of terms, and organization of the report.

Chapter Two reviews relevant literature on Artificial Intelligence, educational service assistants, Natural Language Processing, Machine Learning, educational technologies, theoretical frameworks, previous studies, current trends, comparative analysis of existing approaches, research gaps, and the relevance of AI in higher education.

Chapter Three discusses the proposed AI-Powered Educational Service Assistant in detail. It presents the major components of the system, techniques and models, tools and technologies,

applications, benefits, implementation challenges, case studies, emerging issues, and future directions for AI adoption in Nigerian tertiary institutions.

Chapter Four summarizes the major findings of the study, presents the conclusion, offers recommendations for successful implementation of AI-powered educational service assistants, and suggests areas for further research

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The literature review is an essential component of academic research because it provides a systematic examination of existing knowledge, theories, concepts, and empirical findings relating to the subject under investigation. It enables the researcher to understand what previous scholars have established, identify areas of agreement and disagreement, examine methodological approaches adopted by earlier researchers, and identify gaps that require further investigation. A properly conducted literature review also provides the theoretical and conceptual foundation upon which a study is developed.

The present study focuses on the development of an AI-Powered Educational Service Assistant for tertiary institutions. Consequently, the literature reviewed in this chapter covers Artificial Intelligence (AI), educational service assistants, Natural Language Processing (NLP), Machine Learning (ML), chatbots, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), generative AI, technology acceptance, and the application of AI in higher education.

Artificial Intelligence has become one of the most influential technologies of the twenty-first century. Its applications extend across healthcare, agriculture, banking, manufacturing, transportation, security, business, government, and education. In higher education, AI has been investigated and implemented for intelligent tutoring, automated assessment, student performance prediction, learning personalization, academic advising, administrative support, and institutional management. A systematic review by Zawacki-Richter et al. (2019) demonstrated that AI applications in higher education encompass areas such as assessment and evaluation, adaptive systems, profiling and prediction, and intelligent tutoring systems.

More recently, the development of conversational AI and generative AI has expanded the possible applications of AI in educational institutions. AI-powered chatbots can interact with

students using natural language, answer frequently asked questions, provide explanations, direct students to relevant institutional resources, and support administrative processes. A systematic review of chatbot applications in education found that chatbots have the potential to provide quick and personalized services to students and institutional employees (Okonkwo & Ade-Ibijola, 2021).

The increasing use of AI in education has also created concerns relating to accuracy, privacy, ethical use, bias, transparency, academic integrity, and overdependence on automated systems. UNESCO (2023) therefore advocates a human-centred approach to generative AI in education, emphasizing privacy protection, ethical validation, equity, inclusion, and appropriate institutional policies.

This chapter therefore examines the relevant concepts, theories, empirical studies, current technological developments, existing approaches, and identified gaps that provide the foundation for the development of an AI-powered educational service assistant.

2.2 Conceptual Review

2.2.1 Concept of Artificial Intelligence

Artificial Intelligence is a branch of computer science concerned with the development of computer systems capable of performing tasks that ordinarily require aspects of human intelligence. Such tasks include learning, reasoning, problem-solving, perception, language understanding, pattern recognition, and decision-making. Russell and Norvig (2021) explain AI through the concept of intelligent agents that perceive their environments and take actions aimed at achieving specified objectives.

The development of Artificial Intelligence has involved contributions from several disciplines, including computer science, mathematics, statistics, psychology, linguistics, philosophy, neuroscience, and cognitive science. Modern AI systems are increasingly based on data-driven approaches in which algorithms learn patterns from large datasets and use the learned patterns

to make predictions or generate outputs.

AI is not a single technology. Rather, it represents a broad collection of methods and technologies. These include Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Expert Systems, Robotics, Speech Recognition, and Generative AI. Each of these technologies performs different functions, although they can be combined to develop sophisticated intelligent applications.

In educational institutions, AI can be applied to both academic and administrative activities. Academic applications include intelligent tutoring, personalized learning, automated assessment, recommendation systems, and student performance prediction. Administrative applications include admission support, student enquiries, information retrieval, scheduling, records management, and automated communication.

Zawacki-Richter et al. (2019) noted that AI research in higher education has increasingly focused on applications that can support educational processes and institutional activities.

Characteristics of Artificial Intelligence

Artificial Intelligence possesses several characteristics that distinguish intelligent systems from conventional computer applications. Important characteristics include:

1. **Learning ability:** AI systems can learn patterns from data and improve their performance based on available information.
2. **Pattern recognition:** AI can identify relationships and patterns within large amounts of data that may be difficult for humans to detect manually.
3. **Natural language understanding:** AI systems can process and respond to human language through Natural Language Processing.
4. **Decision-making:** AI systems can analyze information and provide recommendations or make decisions based on predefined rules or learned patterns.
5. **Problem-solving:** AI can be designed to identify solutions to problems by evaluating

available information and possible alternatives.

6. **Automation:** AI can automate repetitive tasks, thereby reducing human workload and improving operational efficiency.
7. **Adaptability:** Some AI systems can adjust their responses when new information becomes available.
8. **Predictive capability:** Machine Learning algorithms can analyze historical information to predict likely future outcomes.
9. **Contextual interaction:** Modern conversational AI systems can use contextual information to produce responses that are more relevant to users' enquiries.

These characteristics make AI particularly suitable for educational service assistants, where students may submit different questions using different forms of natural language.

2.2.2 Concept of Educational Service Assistant

An educational service assistant is a digital system designed to provide information, guidance, and support to students, staff, prospective students, and other stakeholders of an educational institution. Traditionally, educational support services have depended primarily on human personnel working in admission offices, student affairs divisions, academic departments, ICT centres, libraries, bursary departments, and examination offices.

The emergence of digital technologies has changed this traditional approach. Websites, portals, mobile applications, email systems, and social media platforms have enabled students to access some institutional information electronically. However, many conventional digital platforms still require users to search through menus, documents, web pages, and frequently asked questions before locating the information they need.

An AI-powered educational service assistant attempts to overcome some of these limitations by allowing users to communicate with the system conversationally. Instead of searching manually through multiple pages, a student may type a question such as:

"What are the requirements for course registration?"

The system can process the question, identify the user's intention, search its approved knowledge source, and return an appropriate response.

Educational chatbots and conversational agents have increasingly been studied as tools for student and institutional support. Ramandanis and Xinogalos (2023), in a systematic review, examined the support provided by chatbots to educational institutions and students and identified applications involving student support and institutional services.

Examples of services that may be provided by an AI educational service assistant include:

- Admission enquiries.
- Course registration guidance.
- School fee information.
- Academic calendar information.
- Examination timetable enquiries.
- Result-checking guidance.
- Hostel accommodation information.
- SIWES information.
- Library service information.
- Transcript application guidance.
- Graduation requirements.
- Departmental contact information.
- General institutional enquiries.
- Frequently asked questions.

The proposed system in this study is therefore conceptualized as an intelligent communication platform between the institution and its users.

2.2.3 Natural Language Processing (NLP)

Natural Language Processing is a branch of Artificial Intelligence that enables computers to process, interpret, analyze, and generate human language. NLP combines techniques from computer science, linguistics, statistics, and Machine Learning to facilitate communication between humans and computer systems.

NLP is particularly important to an educational service assistant because users are expected to communicate using ordinary language rather than programming commands. For example, a student may ask:

"When will the examination timetable be released?"

The system must be capable of identifying the meaning and intention behind the question before producing an appropriate response.

Major NLP processes include:

Text Pre-processing

Text pre-processing involves preparing textual information for computational analysis. It may involve removing unnecessary characters, correcting formatting problems, converting text to an appropriate format, and preparing the input for further processing.

Tokenization

Tokenization involves dividing a sentence or document into smaller units known as tokens. These may include words, phrases, or sub-word units depending on the NLP model being used.

Part-of-Speech Tagging

Part-of-speech tagging involves identifying the grammatical role of words in a sentence, such as nouns, verbs, adjectives, and adverbs.

Named Entity Recognition

Named Entity Recognition identifies specific entities within text, such as names of institutions, departments, persons, dates, locations, and other important terms.

Intent Recognition

For an educational chatbot, intent recognition is particularly important because the system needs to determine what the user wants. For example, questions about school fees should be classified differently from questions about examination timetables.

Response Generation

After determining the user's intention and retrieving relevant information, the system generates an appropriate response. Modern conversational systems can use large language models to generate natural-sounding responses.

The effectiveness of an AI educational service assistant therefore depends significantly on its ability to interpret user language correctly.

2.2.4 Machine Learning

Machine Learning is a subfield of Artificial Intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions, classifications, or decisions without requiring every possible rule to be explicitly programmed.

Machine Learning is important because educational environments generate substantial amounts of data. Such data may include student records, examination results, registration information, enquiries, attendance records, and other institutional information. Machine Learning techniques can be used to identify patterns within such datasets.

The major categories of Machine Learning include supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

Supervised learning involves training a model using labelled data. The model learns the relationship between input variables and known outputs.

Possible applications in education include:

- Student performance prediction.

- Classification of student enquiries.
- Admission decision support.
- Identification of frequently asked questions.
- Classification of institutional documents.

Unsupervised Learning

Unsupervised learning involves analyzing data without predetermined labels. The objective is often to discover hidden patterns, groups, or relationships.

Possible applications include:

- Grouping similar student enquiries.
- User segmentation.
- Identification of common information needs.
- Analysis of student behaviour.

Reinforcement Learning

Reinforcement learning involves an agent learning through interaction with an environment. The system receives feedback based on its actions and gradually learns strategies that produce better outcomes.

In a conversational educational assistant, user feedback can potentially be used to evaluate the quality of responses and improve the system over time. However, the use of such feedback must be carefully controlled to avoid learning incorrect or inappropriate information.

2.2.5 Large Language Models

Large Language Models are advanced AI models trained on large quantities of textual data to understand and generate human language. They form the technological foundation of many modern conversational AI applications.

LLMs can perform tasks such as:

- Question answering.

- Text generation.
- Summarization.
- Translation.
- Classification.
- Information explanation.
- Conversational interaction.

Their relevance to an educational service assistant lies in their ability to understand differently worded questions. For example, the questions *"How much are the school fees?"*, *"What is the current tuition fee?"*, and *"How much do I need to pay?"* may express a similar underlying intention.

However, LLMs may sometimes generate incorrect information, a phenomenon commonly referred to as hallucination. Consequently, an institutional AI assistant should not rely entirely on unrestricted generation when providing official institutional information.

This is one reason why grounding AI responses in verified institutional information is important.

2.2.6 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is an approach that combines information retrieval with generative AI. Rather than relying solely on information contained within the model, the system first retrieves relevant information from an external knowledge base and then uses that information to generate its response.

For an educational institution, the knowledge base could contain verified documents such as:

- Student handbooks.
- Academic calendars.
- Registration guidelines.
- School fee schedules.

- Admission requirements.
- Examination regulations.
- SIWES guidelines.
- Departmental information.
- Frequently asked questions.
- Institutional policies.

When a student submits a question, the system searches the institutional knowledge base for relevant information. The retrieved information is then provided to the language model to assist in generating the response.

This approach is particularly relevant to the proposed system because institutional information can change over time. A retrieval-based system can allow administrators to update the knowledge base without necessarily retraining the entire language model.

However, RAG does not automatically eliminate inaccurate answers. The quality of the response still depends on the quality, completeness, currency, and authority of the documents retrieved. Appropriate system design, source control, and human oversight are therefore necessary.

2.2.7 Educational Chatbots

An educational chatbot is a conversational software application that uses Artificial Intelligence and related technologies to communicate with users. Chatbots can operate through websites, mobile applications, messaging platforms, or institutional portals.

Educational chatbots may be broadly classified into:

Rule-Based Chatbots

Rule-based chatbots operate according to predefined rules and decision trees. They are suitable for simple and predictable questions but may struggle with questions that are phrased differently from the programmed rules.

AI-Based Chatbots

AI-based chatbots use NLP and Machine Learning to understand user questions and produce responses. They are generally more flexible than traditional rule-based systems.

Generative AI Chatbots

Generative AI chatbots use large language models to generate responses based on user input and available context.

Research indicates that AI chatbots can provide immediate support, explanations, personalized assistance, and access to educational resources. Labadze et al. (2023), after reviewing 67 studies, reported benefits associated with AI chatbots while also highlighting challenges involving reliability, accuracy, and ethical considerations.

A more recent systematic review also emphasizes that chatbots have become an important area of research in education, including chatbot design, student adoption, human-chatbot interaction, and educational applications.

2.3 Historical Background of the Topic

The history of Artificial Intelligence can be traced to early attempts by mathematicians, philosophers, and computer scientists to understand whether machines could perform activities associated with human reasoning.

A major milestone occurred in 1950 when Alan Turing published his influential work on machine intelligence and proposed what later became known as the **Turing Test**. The test provided a conceptual method for considering whether a machine could exhibit behaviour that would be regarded as intelligent.

The term **Artificial Intelligence** was formally introduced in 1956 at the Dartmouth Summer Research Project on Artificial Intelligence. The workshop, associated with researchers including John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon, is widely regarded as an important milestone in the establishment of AI as an academic field.

During the 1960s and 1970s, AI research focused considerably on symbolic reasoning, logic, problem-solving, and expert systems. Early AI programs demonstrated that computers could perform specialized tasks using predefined rules.

During subsequent decades, AI research experienced periods of optimism and reduced funding commonly described as AI winters. Nevertheless, research continued in areas such as expert systems, machine learning, robotics, and natural language processing.

The emergence of the internet and increasing availability of digital data in the 1990s and 2000s contributed to the growth of data-driven Machine Learning. AI became increasingly visible in search engines, recommendation systems, speech recognition, fraud detection, and digital assistants.

The 2010s witnessed major progress in Deep Learning. Neural networks with multiple layers achieved significant improvements in image recognition, speech processing, language translation, and natural language applications.

The development of transformer-based language models subsequently contributed to the rapid growth of conversational AI and generative AI. This development has significantly expanded the possible applications of AI in education.

Today, AI systems are being investigated for teaching, learning, assessment, academic support, student services, and institutional administration. Zawacki-Richter et al. (2019) documented the growing body of research on AI applications in higher education.

The historical development of AI therefore demonstrates a movement from rule-based systems toward data-driven, conversational, and generative systems. This evolution provides the technological foundation for the proposed AI-powered educational service assistant.

2.4 Theoretical Framework

A theoretical framework provides the intellectual foundation for a research study. It identifies theories that explain the relationships among the concepts being investigated and provides a

basis for interpreting the findings.

The present study is concerned not only with the technical development of an AI-powered educational service assistant but also with how potential users may perceive and accept the technology. Consequently, the **Technology Acceptance Model (TAM)** is considered particularly relevant.

2.4.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model was developed by Davis (1989) to explain why individuals accept or reject information technologies. The model proposes that two major perceptions influence technology acceptance: **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEOU)**.

Perceived Usefulness

Perceived usefulness refers to the extent to which a person believes that using a particular technology will improve his or her performance.

In relation to the proposed AI-powered educational service assistant, students are more likely to use the system if they believe it will:

- Provide information quickly.
- Reduce the need to visit administrative offices.
- Help them obtain accurate institutional information.
- Save time.
- Improve access to educational services.
- Provide assistance outside normal office hours.

Perceived Ease of Use

Perceived ease of use refers to the extent to which a person believes that using a technological system will require little effort.

The proposed educational assistant should therefore be easy to use. Students should be able to enter questions naturally without requiring technical knowledge.

Application of TAM to the Study

TAM is relevant to this study because the success of an AI-powered educational service assistant depends not only on its technical capabilities but also on whether students and staff are willing to use it.

For example, a technologically sophisticated chatbot may fail to achieve its intended purpose if students consider it difficult to use. Similarly, students may reject a simple system if they do not believe that it provides useful information.

Davis (1989) established perceived usefulness and perceived ease of use as important determinants of technology acceptance.

Therefore, TAM provides an appropriate theoretical foundation for evaluating the likely acceptance of the proposed AI-powered educational service assistant.

2.5 Review of Related Studies

Study by Zawacki-Richter et al. (2019)

Zawacki-Richter et al. (2019) conducted a systematic review of research on Artificial Intelligence applications in higher education. Their review examined the areas in which AI was being applied within higher education and identified applications involving assessment and evaluation, adaptive systems, profiling and prediction, and intelligent tutoring.

The study demonstrated that AI has considerable potential within higher education. However, the researchers also highlighted the need for greater attention to the role of educators and the pedagogical implications of AI.

The relevance of this study to the present research is that it establishes the broad applicability of AI in higher education. However, the study was not specifically focused on developing an AI-powered administrative service assistant for Nigerian tertiary institutions. This creates room

for the present research.

Study by Okonkwo and Ade-Ibijola (2021)

Okonkwo and Ade-Ibijola (2021) systematically reviewed chatbot applications in education. Their study analyzed 53 articles and examined the benefits, challenges, and potential future applications of chatbot technology.

The researchers found that chatbots could provide quick and personalized services to users in educational environments. However, they also identified challenges associated with implementation, including technical limitations and the need for appropriate design and evaluation.

The study is relevant because the proposed educational service assistant incorporates chatbot functionality. Nevertheless, the review covered chatbot applications broadly and did not specifically address an AI-powered educational service assistant designed around the administrative information needs of a Nigerian polytechnic.

Study by Labadze et al. (2023)

Labadze et al. (2023) conducted a systematic literature review on the role of AI chatbots in education. The researchers analyzed 67 studies and identified several benefits of chatbots, including immediate assistance, personalized learning support, homework assistance, and support for educators.

The study also identified concerns relating to reliability, accuracy, and ethical issues.

The relevance of this study to the present research lies in its demonstration that AI chatbots can improve accessibility to information and provide immediate support. However, the study focuses broadly on education and does not specifically investigate institutional administrative services in Nigerian polytechnics.

Study by Ramandanis and Xinogalos (2023)

Ramandanis and Xinogalos (2023) conducted a systematic review of chatbot support for

educational institutions and students. Their research examined the emerging roles of conversational agents and how these systems could be used to provide educational and institutional services.

The researchers emphasized that chatbots can be used to support students and educational institutions, although the successful implementation of such systems depends on appropriate design and consideration of user needs.

The study supports the idea that an AI-powered educational service assistant can serve as a communication interface between students and institutions. Nevertheless, there remains a need for systems specifically designed around the administrative and informational needs of Nigerian tertiary institutions.

Study by Al-Shamaileh et al. (2026)

Al-Shamaileh et al. (2026) conducted a systematic review of ethical considerations surrounding AI chatbots in higher education. Their review examined 109 studies and identified recurring concerns relating to academic integrity, privacy and security, algorithmic bias, overdependence on automated systems, and inaccurate or misleading outputs.

The study is relevant because an AI-powered educational service assistant will process user enquiries and potentially interact with institutional information. Consequently, issues relating to data protection, reliability, transparency, and human oversight must be considered during system development.

The study further demonstrates that AI implementation in education should not focus exclusively on technological functionality. Ethical and institutional considerations are equally important.

2.6 Current Trends and Technologies

Artificial Intelligence continues to evolve rapidly. New AI technologies have expanded the capabilities of educational applications from simple automated responses to sophisticated

conversational systems.

2.6.1 Generative Artificial Intelligence

Generative AI refers to AI systems capable of producing new content based on user instructions. Generated content may include text, images, audio, video, and computer code.

Examples of widely known generative AI systems include:

- ChatGPT.
- Google Gemini.
- Microsoft Copilot.
- Claude.

In education, generative AI has been applied or investigated for:

- Student support.
- Explanation of difficult concepts.
- Academic writing assistance.
- Content generation.
- Lesson planning.
- Feedback.
- Research assistance.
- Personalized learning.

Recent research indicates that generative AI chatbots can provide immediate feedback and personalized support, although concerns remain regarding accuracy, independent learning, academic integrity, and appropriate use.

UNESCO (2023) emphasizes that generative AI should be introduced through a human-centred approach that considers privacy, equity, ethical use, institutional preparedness, and human agency.

For the proposed system, generative AI can be useful for converting retrieved institutional

information into understandable conversational responses.

2.6.2 Large Language Models

Large Language Models have become central to modern conversational AI. They are capable of processing and generating natural language and can respond to a wide range of user questions.

The application of LLMs to educational service delivery may allow students to communicate with an institutional system using ordinary language. However, because LLMs may generate plausible but incorrect information, institutional applications require reliable knowledge sources and appropriate safeguards.

2.6.3 Retrieval-Augmented Generation

RAG represents an important development for institutional AI systems because it allows responses to be grounded in external information sources.

For the proposed system, RAG can be used to connect the AI model with an institutional database or document repository. When a student asks a question, relevant institutional information can be retrieved before the response is generated.

This approach can help reduce the risk of the system relying solely on general knowledge and can make it easier to update institutional information.

2.6.4 Conversational AI

Conversational AI enables computer systems to interact with humans using natural language. It combines NLP, Machine Learning, language models, and dialogue-management techniques. Conversational AI is particularly suitable for educational service delivery because students are already familiar with messaging and conversational interfaces.

2.7 Comparative Analysis of Existing Approaches

Educational institutions have traditionally used several approaches for delivering information and administrative support. These approaches include face-to-face support, telephone

communication, static websites, frequently asked questions, student portals, and AI-powered assistants.

Approach	Major Strength	Major Limitation
Face-to-face enquiry	Human interaction and clarification	Time-consuming and dependent on staff availability
Telephone support	Direct communication	May involve waiting and limited availability
Notice boards	Simple and inexpensive	Information may become outdated
Institutional website	Provides large amounts of information	Users may struggle to locate specific information
FAQ system	Provides answers to common questions	Limited flexibility in interpreting questions
Student portal	Can provide personalized information	Usually requires navigation and authentication
Rule-based chatbot	Automated and relatively simple	Limited ability to understand varied language
AI-powered assistant	Conversational, scalable and potentially available 24/7	Requires reliable data, safeguards and maintenance

The comparison demonstrates that conventional approaches continue to have value, particularly where human judgement is required. However, they can be complemented by AI-powered systems for routine enquiries.

A conventional website may contain the required information but still require students to navigate multiple pages. An AI assistant, by contrast, can provide a conversational interface through which users can directly ask questions.

Systematic reviews have identified immediate support and accessibility as important advantages of educational chatbots while also emphasizing limitations involving accuracy and reliability.

Therefore, the proposed system should not necessarily be viewed as a replacement for existing institutional services. Instead, it should function as an additional support channel that handles routine enquiries and directs complex cases to appropriate human personnel.

2.8 Identified Research Gaps

Although considerable research has been conducted on Artificial Intelligence in education, several gaps remain.

Gap One: Limited Focus on Administrative Support

A significant proportion of existing AI-in-education research focuses on teaching, learning, intelligent tutoring, adaptive learning, automated assessment, and personalized learning. Zawacki-Richter et al. (2019) demonstrated the breadth of AI research in higher education, but there remains a need for more institution-specific research into AI as a tool for administrative information and student support.

The present study addresses this gap by focusing specifically on an educational service assistant capable of responding to routine institutional enquiries.

Gap Two: Limited Nigerian Context

Much of the existing literature on AI-powered educational assistants originates from institutions outside Nigeria. The technological, administrative, infrastructural, and user contexts of Nigerian tertiary institutions may differ considerably from those of institutions in developed countries.

There is therefore a need for research that considers the specific information needs and operational realities of Nigerian tertiary institutions.

Gap Three: Limited Polytechnic-Specific Research

Existing studies frequently examine universities or education generally. Comparatively less attention has been given to the development of AI-powered administrative support systems specifically for polytechnics.

The present research focuses on The Federal Polytechnic Ado-Ekiti, thereby providing an institution-specific perspective.

Gap Four: Limited Integration of Institutional Knowledge

Many general-purpose AI systems are designed to answer broad questions. However, an educational service assistant requires access to institution-specific information such as registration procedures, fees, academic calendars, SIWES information, departmental contacts, examination procedures, and institutional policies.

The proposed system therefore emphasizes the integration of institutional knowledge into an AI-powered interface.

Gap Five: Accuracy and Reliability

AI-generated responses can sometimes be inaccurate. This is particularly problematic in educational administration because students may make important decisions based on information provided by the system.

Recent research has highlighted reliability, accuracy, privacy, and ethical concerns associated with AI chatbots in education.

The proposed system therefore needs mechanisms for using verified institutional information and directing uncertain or complex enquiries to human administrators.

Gap Six: Human Oversight

AI should not be assumed to replace human administrative personnel. Some student enquiries may involve sensitive circumstances, exceptions to institutional policies, or decisions requiring professional judgement.

UNESCO (2023) emphasizes human-centred approaches to AI in education.

The proposed system therefore positions AI as a support mechanism rather than an unrestricted replacement for human administrators.

Gap Seven: User Acceptance

The technical development of an AI system does not automatically guarantee adoption. Students and staff must perceive the system as useful and easy to use.

The Technology Acceptance Model provides an appropriate basis for examining these factors because perceived usefulness and perceived ease of use are central components of technology acceptance (Davis, 1989).

2.9 Summary of Literature Review

This chapter reviewed literature relating to Artificial Intelligence and its application to educational service delivery. The review established that AI has become an important technology in the transformation of education and that its applications extend beyond teaching and learning to include administrative support, student services, assessment, prediction, and institutional management.

The conceptual review examined Artificial Intelligence, educational service assistants, Natural Language Processing, Machine Learning, Large Language Models, Retrieval-Augmented Generation, conversational AI, and educational chatbots. These technologies provide the technical foundation for developing an intelligent system capable of interacting with students and providing relevant institutional information.

The historical review demonstrated that AI has evolved from early symbolic reasoning and rule-based systems into modern Machine Learning, Deep Learning, conversational AI, and generative AI. This development has created new opportunities for educational institutions to provide intelligent and accessible services.

The theoretical framework examined the Technology Acceptance Model developed by Davis

(1989). The model emphasizes perceived usefulness and perceived ease of use as important factors influencing users' acceptance of technology. The theory is relevant to the present study because the effectiveness of an AI-powered educational service assistant will depend partly on whether students and staff consider the system useful, easy to use, and capable of meeting their information needs.

The review of related studies established that AI chatbots and conversational systems can provide immediate assistance, personalized support, and access to information. However, existing research has also identified challenges involving reliability, accuracy, privacy, ethics, bias, and overdependence on automated systems.

The comparative analysis showed that traditional approaches such as face-to-face enquiries, telephone communication, websites, FAQs, and student portals remain useful but have limitations relating to accessibility, response time, scalability, and user convenience. AI-powered educational assistants can complement these approaches by providing a conversational and potentially continuously available channel for routine enquiries.

Finally, the literature review identified important gaps, particularly the limited focus on AI-powered administrative support within Nigerian tertiary institutions and the limited attention given specifically to polytechnic environments. These gaps provide justification for the development of an AI-Powered Educational Service Assistant for The Federal Polytechnic Ado-Ekiti.

CHAPTER THREE

DISCUSSION OF THE TOPIC

3.1 Introduction

Artificial Intelligence (AI) has become a transformative technology that is reshaping the way educational institutions deliver academic and administrative services. The increasing demand for efficient communication, timely dissemination of information, and improved student support has encouraged higher institutions around the world to adopt intelligent digital solutions. AI-powered educational service assistants are among the most promising innovations because they enable institutions to automate repetitive administrative tasks, provide real-time responses to enquiries, and enhance the overall educational experience.

In Nigerian tertiary institutions, administrative processes such as admission enquiries, course registration, school fee payment, hostel allocation, examination scheduling, result checking, library services, and student support often involve manual procedures that consume considerable time and resources. These challenges affect students, academic staff, administrative personnel, and prospective applicants, leading to delays, misinformation, and reduced institutional efficiency.

3.2 Major Components of the Topic

The proposed AI-Powered Educational Service Assistant consists of several interconnected components that work together to provide accurate, reliable, and efficient educational support services. Each component performs a specific function that contributes to the overall effectiveness of the system.

3.2.1 User Interface (UI)

The User Interface is the point of interaction between users and the AI assistant. It enables students, staff, and prospective applicants to communicate with the system using text or voice commands.

The interface should be simple, intuitive, and user-friendly to accommodate users with different levels of digital literacy. It can be accessed through:

- Institutional website
- Mobile application
- Student portal

3.3 Techniques, Methods, or Models

The proposed AI-Powered Educational Service Assistant utilizes several Artificial Intelligence techniques and computational models to provide accurate and intelligent services.

3.3.1 Natural Language Processing (NLP)

Natural Language Processing enables the system to understand human language.

Major NLP techniques include:

- Tokenization
- Stemming
- Lemmatization
- Intent classification
- Named Entity Recognition
- Sentiment Analysis

3.4 Tools and Technologies Used

Developing an AI-powered educational assistant requires integrating several software and hardware technologies.

3.4.1 Programming Languages

Common programming languages include:

- Python
- JavaScript
- Java

- PHP

3.5 Applications and Use Cases

Artificial Intelligence (AI)-powered educational service assistants have a wide range of applications within tertiary institutions. By integrating Natural Language Processing (NLP), Machine Learning (ML), Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG), the proposed system can provide intelligent, efficient, and personalized support to students, academic staff, administrative personnel, parents, and prospective applicants. This section discusses the major applications and use cases of the proposed AI-powered educational service assistant for **The Federal Polytechnic Ado-Ekiti**.

3.5.1 Admission Support

Admission is one of the busiest periods in any tertiary institution. Thousands of prospective students make enquiries regarding admission requirements, application procedures, screening exercises, departmental cut-off marks, required documents, and deadlines.

The AI assistant can automate these enquiries by providing:

- Admission requirements for ND and HND programmes.
- Available courses and departments.
- JAMB subject combinations.
- Post-UTME screening guidelines.

3.6 Benefits and Advantages

The implementation of an AI-powered educational service assistant offers numerous benefits to students, academic staff, administrators, and institutional management.

3.6.1 Twenty-Four-Hour Service Availability

Unlike human administrative staff who operate during official working hours, the AI assistant remains available twenty-four hours a day and seven days a week.

Students can obtain assistance:

- During weekends.
- At night.
- During holidays.
- Outside office hours.

3.7 Challenges and Limitations

Despite its numerous benefits, implementing an AI-powered educational service assistant presents several challenges.

3.7.1 Inadequate ICT Infrastructure

Many Nigerian tertiary institutions experience:

- Poor internet connectivity.
- Limited bandwidth.
- Obsolete computer systems.
- Inadequate server capacity.

3.8 Case Studies

Case studies provide practical examples of how Artificial Intelligence (AI) has been applied to improve educational services and administrative operations in higher institutions. They demonstrate the effectiveness of AI-powered educational assistants in solving real-world problems and provide valuable lessons that can guide the implementation of similar systems in Nigerian tertiary institutions. This section examines selected international and Nigerian case studies relevant to the proposed AI-Powered Educational Service Assistant for **The Federal Polytechnic Ado-Ekiti**.

3.8.1 Case Study One: Georgia State University (United States)

Georgia State University introduced an AI-powered chatbot named **Pounce** to assist prospective and newly admitted students. The chatbot was developed to answer questions relating to admissions, financial aid, registration, orientation, and enrolment procedures.

Features of the System

24-hour student support.

Automated responses to frequently asked questions.

SMS-based communication.

3.9 Emerging Issues

Artificial Intelligence continues to evolve rapidly, introducing new opportunities and challenges for educational institutions. As AI technologies become more sophisticated, several emerging issues require careful consideration.

3.9.1 Responsible Artificial Intelligence

Responsible AI emphasizes that intelligent systems should be:

- Fair.
- Transparent.
- Explainable.
- Accountable.
- Inclusive.
- Safe.

Educational institutions must ensure that AI systems make decisions ethically while protecting users' rights.

3.10 Future Directions

Artificial Intelligence continues to evolve rapidly, creating exciting opportunities for educational institutions.

Several future developments are expected to improve AI-powered educational service assistants.

CHAPTER FOUR

DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter presents the design and implementation of the AI-Powered Educational Service Assistant developed to provide students and other users with quick access to academic and administrative information. The system is designed to provide an intelligent conversational platform through which users can make enquiries concerning admission, course registration, school fees, examinations, SIWES, and other institutional services.

The system combines a chatbot interface with Artificial Intelligence techniques, Natural Language Processing (NLP), a knowledge database, and a response-generation mechanism. The objective is to reduce the difficulties associated with obtaining routine institutional information through physical visits to administrative offices.

The chapter discusses the system design, system architecture, functional modules, database design, input and output design, implementation technologies, system interfaces, testing procedures, and results obtained from the developed prototype.

4.2 System Design

The proposed system was designed as a web-based AI educational service assistant. The design allows users to communicate with the system through a simple chatbot interface.

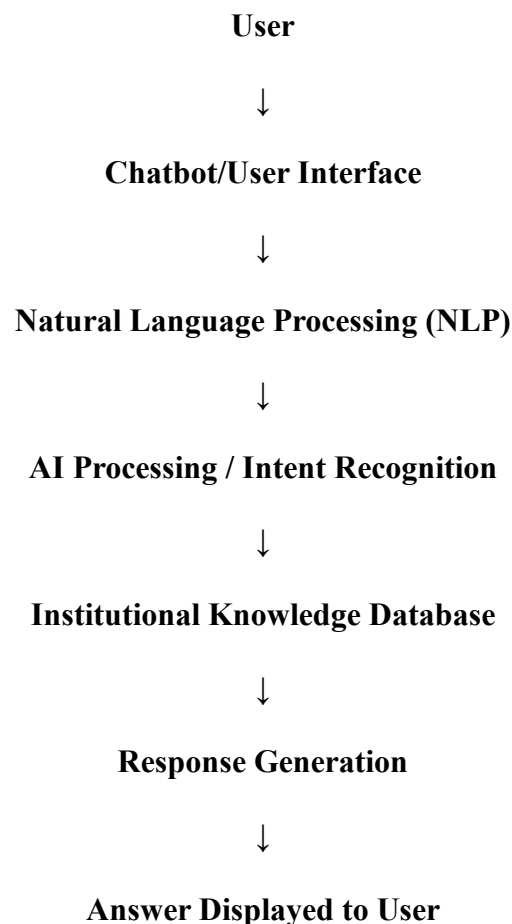
The major components of the system are:

1. **User Interface**
2. **Chatbot Interface**
3. **Natural Language Processing Module**
4. **AI Processing Module**
5. **Knowledge Database**
6. **Response Generation Module**

7. Administrative/Knowledge Management Module

The system receives a user's question through the chatbot interface. The question is processed by the NLP component, after which the system determines the user's intention and searches the relevant institutional knowledge. The appropriate information is then processed and returned to the user as a conversational response.

Figure 4.1: General System Architecture



This architecture allows the system to function as an intermediary between students and institutional information.

4.3 Description of the System Components

4.3.1 User Interface

The user interface provides the point of interaction between the user and the system. It is designed to be simple and user-friendly so that students do not require specialized computer

knowledge to use the application.

The interface contains a chatbot window where users can type questions and receive responses.

4.3.2 Chatbot Module

The chatbot module manages conversations between users and the AI system.

A user may ask questions such as:

- "What are the requirements for admission?"
- "How do I register my courses?"
- "When will examinations begin?"
- "How much is the school fee?"
- "What is SIWES?"
- "How can I apply for hostel accommodation?"

The chatbot receives the question and passes it to the NLP/AI processing component.

4.3.3 Natural Language Processing Module

The NLP module interprets the user's input. It identifies important words and determines the likely intention of the enquiry.

For example:

User	input:
-------------	---------------

"How do I register my courses?"

Detected	intention:
-----------------	-------------------

Course Registration

The system then searches the relevant knowledge source for information associated with course registration.

4.3.4 AI Processing Module

The AI processing module determines how the system should respond to a user's enquiry.

It considers the user's question, detected intent, available institutional information, and

conversation context before producing the response.

The AI component allows users to ask questions in different ways while still obtaining relevant information.

For example:

- "How can I register my courses?"
- "What's the procedure for course registration?"
- "I want to register my courses, what should I do?"

Although the wording differs, the system should recognize that all three questions relate to course registration.

4.3.5 Knowledge Database

The knowledge database stores information required by the educational service assistant.

The database may contain information relating to:

Category	Information Stored
Admission	Requirements and procedures
Registration	Course registration procedures
School Fees	Payment guidance and fee information
Examination	Examination schedules and guidelines
SIWES	SIWES procedures and requirements
Library	Library information
Hostel	Accommodation information
Graduation	Graduation requirements
General Information	Frequently asked questions

The knowledge base should be regularly updated to ensure that users receive current institutional information.

4.3.6 Response Generation Module

After processing the user's enquiry, the system generates an appropriate response.

For example:

User:

"How do I register my courses?"

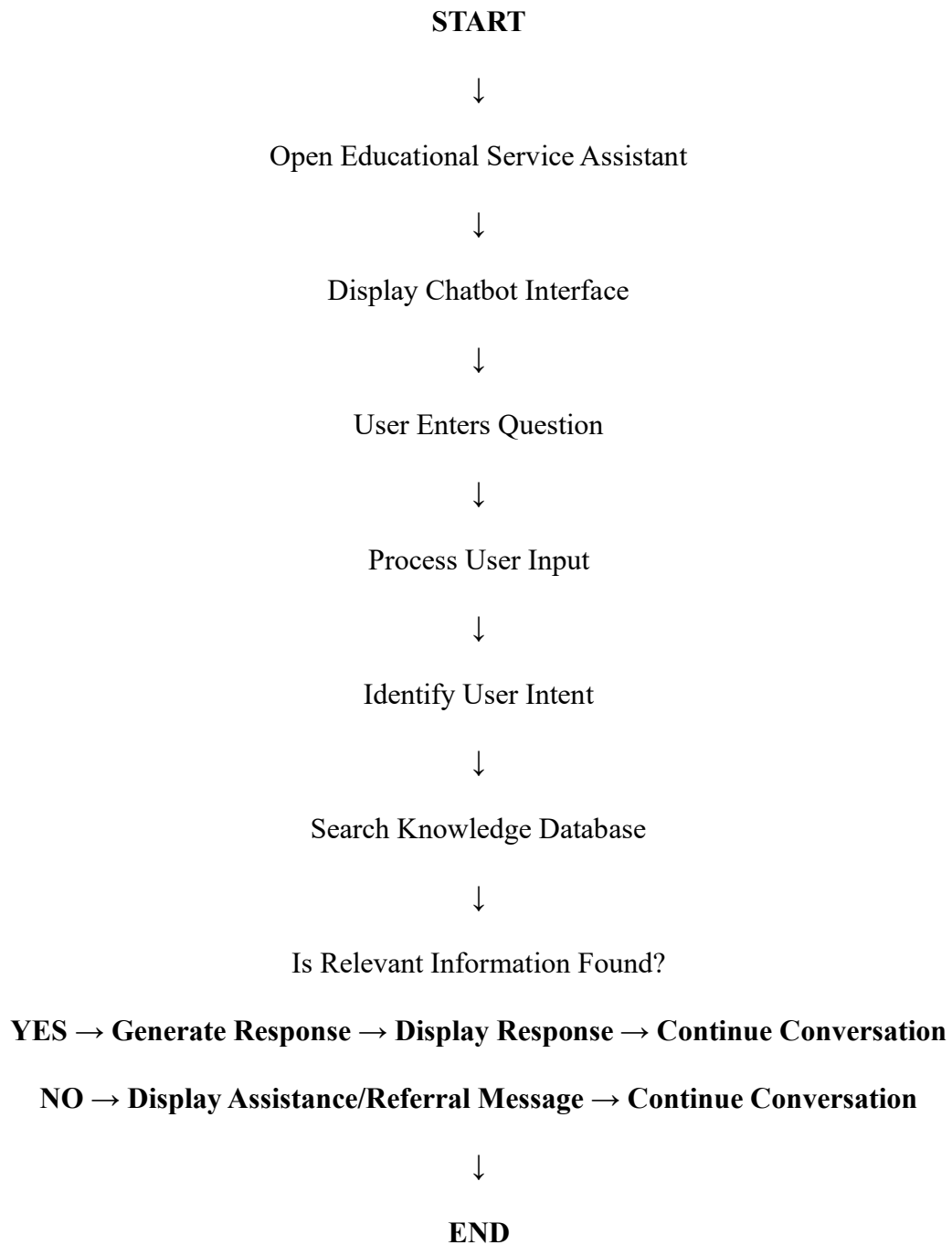
System:

"Course registration is completed through the student portal. Log in with your student credentials, select course registration, choose the appropriate courses for your programme and level, and submit the registration. Contact your department if you experience difficulties."

The response-generation component is responsible for presenting information in a clear and understandable form.

4.4 System Flowchart

The operational flow of the proposed system can be described as follows:



The flowchart ensures that every enquiry passes through a defined processing sequence before a response is provided.

4.5 Functional Modules of the System

4.5.1 Admission Module

This module provides information concerning admission-related enquiries.

It can provide information such as:

- Admission requirements
- Application procedures
- Screening information
- Admission status guidance
- Required documents
- Registration after admission

4.5.2 Course Registration Module

The course-registration module provides information about the process of registering courses.

It can answer questions concerning:

- Course registration procedures
- Course forms
- Registration deadlines
- Course units
- Departmental registration guidance

4.5.3 School Fee Module

This module provides guidance concerning school-fee-related enquiries.

Users can ask questions concerning:

- School fees
- Payment procedures
- Payment deadlines

- Payment confirmation
- Receipts
- Outstanding payments

4.5.4 Examination Module

The examination module provides information about examinations.

It may provide:

- Examination timetable information
- Examination rules
- Examination requirements
- Examination venues
- Result-related guidance

4.5.5 SIWES Module

The SIWES module provides information relating to the Students Industrial Work Experience Scheme.

It can provide guidance concerning:

- SIWES requirements
- Placement procedures
- SIWES documentation
- Logbook information
- Industrial training procedures
- Submission requirements

4.5.6 General Information Module

This module handles general institutional enquiries that do not fall under the specialized categories.

Examples include:

- Library services
- Hostel information
- Departmental information
- Academic calendar
- Contact information
- Graduation information
- General frequently asked questions

4.6 Database Design

The database is designed to store institutional information required by the AI assistant.

A simplified database structure may contain the following tables.

Table 4.1: Users Table

Field	Data Type	Description
user_id	INT	Unique user identification
name	VARCHAR	User's name
email	VARCHAR	User email
password	VARCHAR	User password
role	VARCHAR	User category

Table 4.2: Knowledge Base Table

Field	Data Type	Description
knowledge_id	INT	Unique identification
category	VARCHAR	Information category
question	TEXT	Common question
answer	TEXT	Approved response
date_updated	DATE	Last update

Table 4.3: Chat History Table

Field	Data Type	Description
chat_id	INT	Unique chat identification
user_id	INT	User identification
user_question	TEXT	Question submitted
system_response	TEXT	AI response
date_time	DATETIME	Time of interaction

4.7 Input Design

The major input to the system is the user's enquiry.

The chatbot provides a text input field through which the user enters a question.

Examples of inputs include:

"What are the admission requirements?"

"How can I register my courses?"

"How do I pay my school fees?"

"When is the examination?"

"What documents do I need for SIWES?"

The system is designed to accept natural-language questions rather than requiring users to select complicated commands.

4.8 Output Design

The output of the system is the response generated for the user's enquiry.

The response should be:

- Relevant
- Clear
- Concise

- Understandable
- Helpful
- Based on available institutional information

Where the system cannot confidently answer a question, it should inform the user and recommend contacting the appropriate institutional office rather than presenting uncertain information as fact.

4.9 System Implementation

The system can be implemented using a combination of web development technologies and AI technologies.

Table 4.4: Implementation Technologies

Technology	Purpose
HTML	Structure of web pages
CSS	Interface styling
JavaScript	Client-side interaction
Python	AI/backend programming
Flask/Django	Web application framework
NLP tools	Language processing
AI/LLM service	Conversational response generation
MySQL/PostgreSQL	Database management
Git	Version control

The frontend provides the visual interface, while the backend manages user requests, AI processing, database operations, and response delivery.

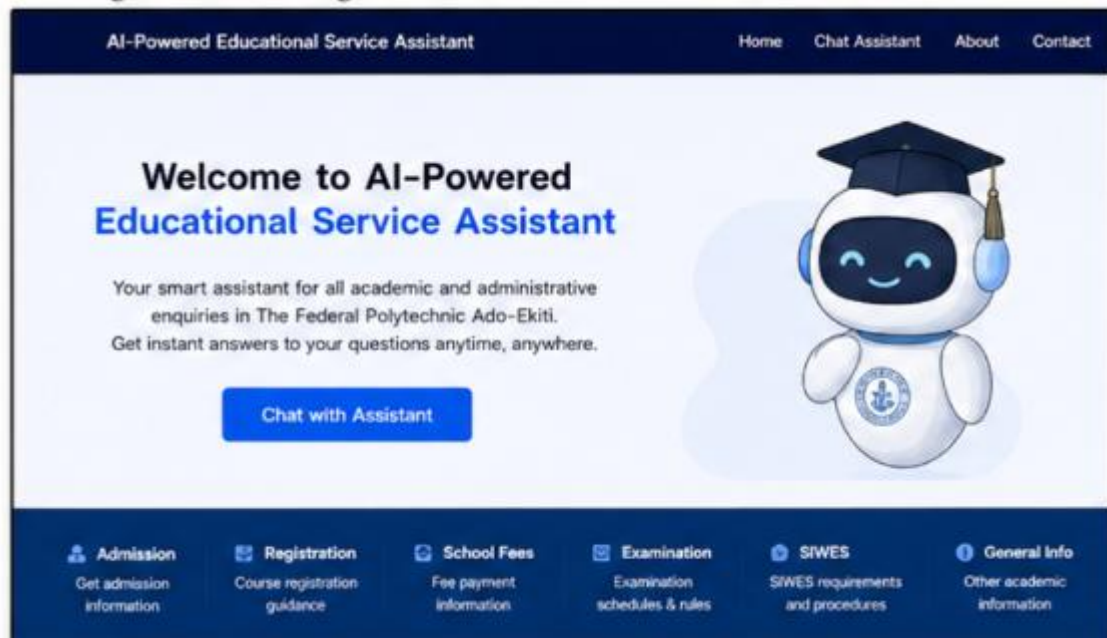
4.10 System Interface

4.10.1 Home Page

The home page provides users with access to the educational service assistant.

It contains the name of the system, a brief description, navigation options, and a button for accessing the chatbot.

Figure 4.1: Home Page of the AI-Powered Educational Service Assistant



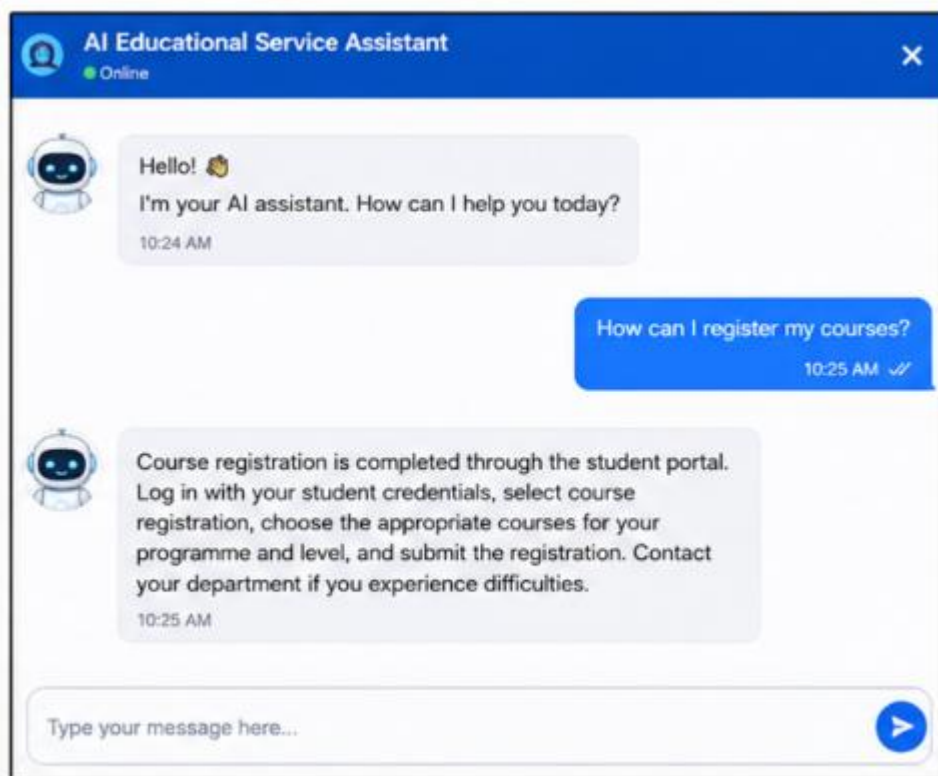
4.10.2 Chatbot Interface

The chatbot interface is the main component through which students interact with the system.

It contains:

- Welcome message
- Chat history
- User input field
- Send button
- AI response area

Figure 4.2: Chatbot Interface



4.10.3 Admission Enquiry Result

The system can process admission-related questions and provide appropriate responses from the institutional knowledge base.

Example interaction:

User:

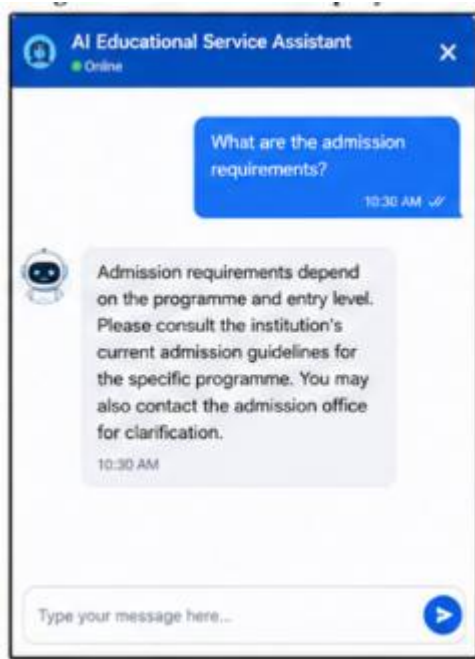
"What are the admission requirements?"

AI

Assistant:

"Admission requirements depend on the programme and entry level. Please consult the institution's current admission guidelines for the specific programme. You may also contact the admission office for clarification."

Figure 4.3: Admission Enquiry Response



4.10.4 Course Registration Result

User:

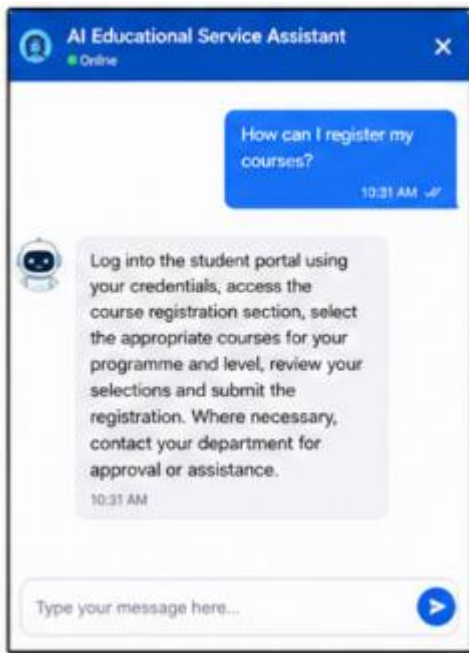
"How can I register my courses?"

AI

Assistant:

"Log into the student portal using your credentials, access the course registration section, select the appropriate courses for your programme and level, review your selections and submit the registration. Where necessary, contact your department for approval or assistance."

Figure 4.4: Course Registration Response



4.10.5 School Fee Enquiry Result

User:

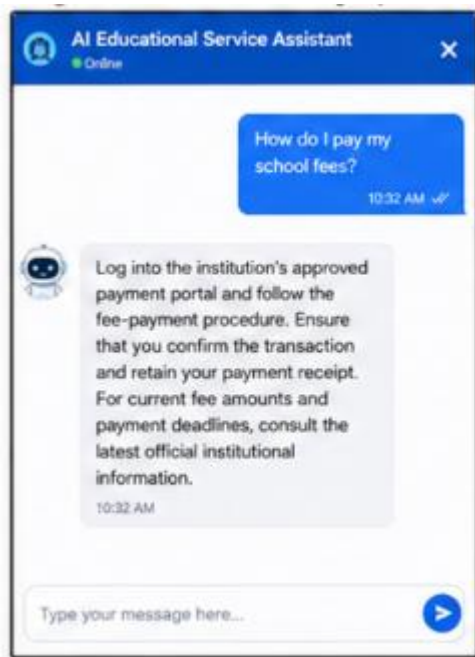
"How do I pay my school fees?"

AI

Assistant:

"Log into the institution's approved payment portal and follow the fee-payment procedure. Ensure that you confirm the transaction and retain your payment receipt. For current fee amounts and payment deadlines, consult the latest official institutional information."

Figure 4.5: School Fee Enquiry Response



4.10.6 Examination Enquiry Result

User:

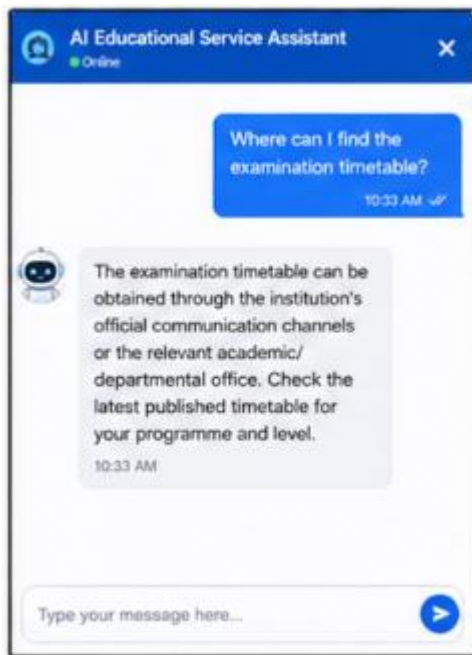
"Where can I find the examination timetable?"

AI

Assistant:

"The examination timetable can be obtained through the institution's official communication channels or the relevant academic/departmental office. Check the latest published timetable for your programme and level."

Figure 4.6: Examination Enquiry Response



4.10.7 SIWES Enquiry Result

User:

"What is required for SIWES?"

AI

Assistant:

"SIWES requirements may include placement documentation, logbook, industrial training information and other documents specified by your department. Students should obtain the current requirements from their department or SIWES unit."

Figure 4.7: SIWES Enquiry Response



4.11 System Testing

Testing was carried out to determine whether the proposed system performs its intended functions correctly.

The major testing areas include:

1. User interface testing.
2. Chatbot input testing.
3. NLP/intent recognition testing.
4. Database retrieval testing.
5. Response-generation testing.
6. Error handling.
7. General usability testing.

Table 4.5: Sample System Test Results

Test	User Input	Expected Result	System Result	Status
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No.				
1	Admission requirements	Admission information displayed	Information displayed	Pass
2	Course registration procedure	Registration guidance displayed	Guidance displayed	Pass
3	School fee enquiry	Fee/payment guidance displayed	Guidance displayed	Pass
4	Examination enquiry	Examination information displayed	Information displayed	Pass
5	SIWES enquiry	SIWES information displayed	Information displayed	Pass
6	Unknown question	Appropriate fallback message	Fallback displayed	Pass
7	Empty input	Input warning displayed	Warning displayed	Pass

4.12 Result of the System

The developed prototype demonstrates that an AI-powered educational service assistant can provide a conversational channel for accessing institutional information.

The system is capable of receiving natural-language questions and returning responses relating to different educational service categories. The prototype also demonstrates how an institutional knowledge base can be integrated with an AI conversational interface.

The major observed outcomes include:

- Faster access to routine information.
- Reduction in the need for repeated physical enquiries.

- Conversational interaction with users.
- Centralized access to educational service information.
- Availability of automated responses.
- Improved convenience for students.
- Ability to handle frequently asked questions.
- Provision for directing complex enquiries to human administrators.

The figures presented in Sections 4.10.1–4.10.7 should be populated with screenshots from the actual implemented prototype. If the system has not yet been run, they should be labelled as *prototype/interface designs* rather than actual test results.

4.13 Discussion of Results

The implementation demonstrates the practical application of Artificial Intelligence to educational service delivery. The chatbot provides an alternative channel through which students can obtain routine information without necessarily visiting administrative offices.

The use of Natural Language Processing allows users to communicate with the system using ordinary language. This is important because students may phrase the same enquiry in several different ways.

The integration of a knowledge database also provides a mechanism for storing institutional information in a structured manner. This makes it possible to update information as institutional procedures, deadlines, fees, and policies change.

The prototype further demonstrates the potential of AI to complement conventional student-support services. Rather than replacing academic or administrative personnel, the system can handle routine enquiries and refer complicated or sensitive cases to the appropriate human personnel.

This approach is consistent with the literature reviewed in Chapter Two, which indicates that AI and chatbot technologies can support higher education while concerns regarding accuracy,

ethics, privacy, and human oversight remain important considerations.

4.14 Summary

This chapter presented the design and implementation of the AI-Powered Educational Service Assistant. The system architecture, components, functional modules, database design, input and output design, implementation technologies, system interfaces, testing procedures, and results were discussed.

The system was designed to provide students with accessible information concerning admission, course registration, school fees, examinations, SIWES, and other educational services.

The implementation demonstrates the potential of Artificial Intelligence, Natural Language Processing, and database technologies to improve the delivery of routine educational support services in tertiary institutions.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary

This study focused on the development and application of an AI-Powered Educational Service Assistant for Nigerian Tertiary Institutions, using The Federal Polytechnic Ado-Ekiti as the case study. The research was motivated by the increasing demand for efficient, reliable, and technology-driven educational support services capable of addressing the administrative challenges commonly experienced by students, staff, and prospective applicants in Nigerian higher institutions.

The study began by examining the background of Artificial Intelligence (AI) and its growing significance in educational administration. It established that although many Nigerian tertiary institutions have adopted Information and Communication Technology (ICT) for online registration, admission processing, fee payment, and result management, many administrative support services still depend on manual procedures. These conventional methods often result in delayed responses to enquiries, excessive workload for administrative personnel, inconsistent dissemination of information, and reduced satisfaction among students and other stakeholders.

5.2 Conclusion

Artificial Intelligence has emerged as one of the most significant technological innovations influencing education across the world. Its ability to simulate human intelligence, understand natural language, analyze data, and automate repetitive tasks has transformed both academic and administrative processes within educational institutions.

This study concludes that the implementation of an AI-Powered Educational Service Assistant would significantly improve educational service delivery within Nigerian tertiary institutions, particularly at The Federal Polytechnic Ado-Ekiti. The proposed system provides an effective

solution to many of the administrative challenges currently experienced by students, lecturers, administrators, and prospective applicants.

5.3 Recommendations

Based on the findings of this study, the following recommendations are made:

1. **Adoption of AI-Powered Educational Assistants:** Nigerian tertiary institutions should adopt AI-powered educational service assistants to improve the efficiency of administrative services and enhance students' access to institutional information.
2. **Investment in ICT Infrastructure:** Institutions should invest in modern ICT infrastructure, including reliable internet connectivity, cloud computing platforms, secure servers, and high-performance computer systems to support AI deployment.
3. **Development of Institutional Knowledge Bases:** Institutions should establish and regularly update comprehensive digital knowledge bases containing verified information on admissions, registration, examinations, fees, academic calendars, regulations, and other student services.
4. **Capacity Building and Training:** Administrative staff, lecturers, and ICT personnel should receive continuous training on Artificial Intelligence technologies, digital literacy, system management, and AI governance to facilitate effective implementation and maintenance.
5. **Data Privacy and Cybersecurity:** Institutions should develop strong data protection policies, implement encryption, multi-factor authentication, role-based access control, and regular security audits to safeguard sensitive institutional information.

5.4 Suggestions for Further Studies

Artificial Intelligence is a rapidly evolving field, and there are several areas requiring further investigation. Future researchers may consider the following topics:

1. Development and implementation of a fully functional AI-powered educational service assistant for **The Federal Polytechnic Ado-Ekiti**.
2. Comparative evaluation of AI-powered educational assistants and traditional student support systems in Nigerian tertiary institutions.
3. Assessment of students' acceptance and perception of Artificial Intelligence in educational administration using the Technology Acceptance Model (TAM).
4. Investigation of the impact of AI-powered educational assistants on student satisfaction, institutional efficiency, and academic performance.
5. Integration of multilingual conversational AI capable of supporting English, Yoruba, Hausa, Igbo, and Nigerian Pidgin within educational institutions.
6. Application of voice-enabled AI assistants for improving accessibility among students with visual impairments and other disabilities.

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